

Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

Team TEXTURA

Weekly Project Progress Report – Week 3

Course: Digital Image Processing

Repository link:

https://github.com/AditiKapadiya/DIP_2026_Project17_Group8.git

Team Members

Aditi Kapadiya	AU2440139
Disha Gadhiya	AU2440217
Khushi Panchal	AU2440002
Harshita Dhanwani	AU2440083

Work Done This Week

- Used multi-radii uniform Local Binary Patterns (LBP) at radius 1, 2 and 3, and normalized the histogram for each radius and assembled them into a feature vector.
- Constructed a “normal texture” reference model for each of the fabrics by taking LBP histograms on nonoverlapping windows of the defect-free patches from Week 2 and averaging them over each fabric type.
- Added a variance filter to remove flat/blank windows (such as background or border areas that have no real fabric texture to bias "normal texture" statistics.
- Used chi-square distance function to compare the distance between a window's LBP histogram and the reference histogram of its fabric type.
- Developed a sliding window deviation map generator that scores each region of a patch to the reference, and creates a full resolution texture-deviation map.

- Applied the Otsu thresholding to the deviation map and obtained a binary predicted defect mask.
- Developed a basic confidence/reliability measure in terms of the difference between the separation of the predicted defect region and background in the deviation map.
- Realized Pixel-level evaluation (IoU, Precision, Recall, F1) between predicted masks and the Week 2 ground-truth masks.
- Applied the entire pipeline to a random sample of patches containing defects and plotted the patch and deviations next to a predicted mask overlay and the ground truth side by side.
- Addressed a scale-mismatch problem where the reference histograms were created using full patches and the small windows used for scoring would have introduced a bias into each deviation score.
- Visually compared the sample outputs and saw clear over-segmentation: The defect regions predicted by the model were much larger and more diverse than the actual defect areas which are thin and concentrated in the ground-truth masks. It is not the LBP/chi-square scoring itself that is the main failure of the current baseline, but rather the thresholding method used, Otsu.

Challenges Faced

- The normal-texture reference was originally generated from 256×256 patches and the deviation was scored from 32×32 windows, which would have yielded systematically biased (overestimation) deviation scores. This was detected and fixed when the reference is built utilizing the same window scale as that used for scoring.
- Contaminated "defect-free" data: some defect-free patches were actually blank/border sections of the background or borders of fabric rather than true fabric texture. It was necessary to identify these and remove them from the data prior to constructing the reference model as they would otherwise skew the "normal texture" statistics.
- The first sampling method, called "Non-representative sampling," took the first defect patches in the order they were discovered, resulting in a small number of source images having the majority of the defects. This was replaced by a sample with a seed value to generate a more representative rough value.
- Computing LBP histograms over many windows that overlap in the patch, and computing them for thousands of defect-free patches, was more expensive than processing at the patch level and had to be lightweight enough to run in Colab.

- One drawback of this fixed single-scale baseline is that Otsu thresholding always sets a split point even when the deviation map is relatively uniform on a relatively defect-free region, which can result in small regions of false-positive deviations.
- Sample patches that were over-segmented on the visual sample: on some sample patches, the defect mask predicted was very large and distributed across the entire patch, but the ground truth defect was a thin streak or a few small pixels. The map of the deviation itself lacks a sharp peak at the actual defect location, and the automatic split by Otsu detects "high" values over most of the patch with an over-prediction. This is a confirmation that thresholding – and not the LBP/chi-square scoring – is the primary cause of lack of precision in the current baseline and is the first thing to fix before the multi-scale/adaptive methods.

Work to be done next week

- Use a second texture descriptor: GLCM (Haralick features – contrast, energy, homogeneity, correlation).
- Apply a Gabor filter bank in various orientations and frequencies and generate energy maps of filter responses.
- Feature vectors of per-window combination of GLCM and Gabor features.
- Evaluate performance of LBP, GLCM and Gabor on a validation subset.
- Choose the top single scale descriptor to use as a baseline for multi-scale and adaptive experiments.
- Implement a more severe thresholding technique (e.g. percentile thresholding or morphological opening followed by Otsu) to be able to filter out the over-segmentation you noticed this week and compare with plain Otsu.